

RHCE – PRACTICAL QUESTION BANK

Maximum Possibilities • 1, 3, 5 and 6 Marks
Topics covered up to Apache HTTPD • NFS excluded

Instructions: Perform the required task on a RHEL-based Linux system. Write the commands used and verify the result wherever applicable.

PART A – 1 MARK PRACTICAL QUESTIONS

1. Write the command to display the current working directory.
2. Write the command to list all files, including hidden files.
3. Write the command to create a directory named lab.
4. Write the command to remove an empty directory.
5. Write the command to create an empty file.
6. Write the command to display the contents of a file.
7. Write the command to display the last 10 lines of a file.
8. Write the command to search for a word inside a file.
9. Write the command to count lines, words and characters in a file.
10. Write the command to display disk-space usage of file systems.
11. Write the command to identify the location of an executable command.
12. Write the command to display memory usage.
13. Write the command to display the system uptime.
14. Write the command to display IP address information.
15. Write the command to test connectivity to another host.
16. Write the command to display the Linux kernel version.
17. Write the command to display CPU information.
18. Write the command to view previously executed commands.
19. Write the command to download a file using wget.
20. Write the command to create a TAR archive.
21. Write the command to extract a TAR archive.
22. Write the command to compress a file using gzip.
23. Write the command to create a hard link.
24. Write the command to create a symbolic link.
25. Write the command to search for a file using find.
26. Write the command to search for a file using locate.
27. Write the command to install a package using DNF.
28. Write the command to remove a package using DNF.
29. Write the command to search for a package using DNF.
30. Write the command to display DNF transaction history.

31. Write the command to create a new user.
32. Write the command to create a new group.
33. Write the command to add a user to a supplementary group.
34. Write the command to display a user's UID and groups.
35. Write the command to delete a user.
36. Write the command to change a user's password.
37. Write the command to change file ownership.
38. Write the command to change file permissions numerically.
39. Write the command to display ACL information.
40. Write the command to set an ACL for a user.
41. Write the command to create a cron entry.
42. Write the command to list a user's cron jobs.
43. Write the command to start a systemd service.
44. Write the command to stop a systemd service.
45. Write the command to enable a service at boot.
46. Write the command to disable a service at boot.
47. Write the command to reload firewalld.
48. Write the command to open a port permanently in firewalld.
49. Write the command to allow an HTTP service through firewalld.
50. Write the command to generate an SSH key pair.
51. Write the command to copy an SSH public key to a remote system.
52. State the default SSH port.
53. Name the main Apache HTTPD service.
54. Write the command to install Apache HTTPD.
55. Write the command to start Apache HTTPD.
56. Write the command to enable Apache HTTPD at boot.

PART B – 3 MARK PRACTICAL QUESTIONS

1. Create a directory structure /project/docs/reports and verify it using ls.
2. Create three files, copy one, move one and delete one using appropriate commands.
3. Create a file and demonstrate the use of cat, head and tail on it.
4. Create a file and use grep and wc to search and count information.
5. Create a TAR archive containing two directories and extract it into another location.
6. Create a gzip-compressed archive and verify its contents.
7. Create a hard link and a symbolic link to a file. Verify both links.
8. Install a package using DNF and verify that it is installed.
9. Search for a package using DNF and display package information.
10. Create a user, create a group, and add the user to the group.

11. Create a user with a specified home directory and verify the account.
12. Modify an existing user's supplementary group membership.
13. Set ownership and group ownership on a directory.
14. Apply read/write/execute permissions to a file using numeric chmod.
15. Apply permissions using symbolic chmod notation.
16. Configure umask and demonstrate its effect on newly created files.
17. Set the sticky bit on a shared directory and verify the permissions.
18. Create an ACL giving a named user additional access to a file.
19. Display and verify ACL entries using getfacl.
20. Configure a network connection using nmcli and verify the IP address.
21. Display NetworkManager connection profiles and identify the active profile.
22. Create a background process and demonstrate jobs, bg and fg.
23. Display running processes using ps and identify a selected process.
24. Change process priority using nice/renice and verify it.
25. Start, stop, enable and disable a systemd service using systemctl.
26. Check the status of a systemd service and verify whether it is enabled.
27. Open a TCP port permanently using firewall-cmd and reload the firewall.
28. Allow HTTP service through firewalld and verify the active rules.
29. Create a cron job that runs a command periodically and verify the crontab.
30. Generate an SSH key pair and configure key-based login to a remote host.
31. Change the SSH server port in sshd_config and restart the SSH service.
32. Install Apache HTTPD, start the service and enable it at boot.
33. Configure the firewall for HTTP and verify Apache from a browser/client.
34. Create a simple web page for Apache and verify that it is served.

PART C – 5 MARK PRACTICAL QUESTIONS

1. Create a new user and group, assign the user to the group, set a password, configure account information and verify the result.
2. Create a shared directory for a group. Configure ownership, group permissions, setgid/sticky-bit requirements and verify access.
3. Create a directory and configure its permissions using both symbolic and numeric chmod. Demonstrate the effect of umask.
4. Create a file with restricted permissions and use ACL to provide access to one additional user. Verify using getfacl.
5. Create a TAR archive of a directory, compress it using gzip, copy it to another location and extract it.
6. Create both hard and symbolic links to a file. Demonstrate their inode/link behavior and verify them using ls.
7. Use DNF to search for a package, display package information, install it, verify the installation, and remove it.

8. Configure a static IPv4 address on a NetworkManager connection using nmcli. Bring the connection up and verify connectivity.
9. Identify a running process, display its details, change its priority, and terminate it safely.
10. Demonstrate foreground and background process management using jobs, bg and fg, and terminate a selected process.
11. Manage a systemd service: start it, stop it, check status, enable it at boot, disable it, and verify each state.
12. Configure firewalld to allow a required TCP port permanently. Reload the firewall and verify the rule.
13. Configure firewalld to allow HTTP service permanently and verify that the service is accessible.
14. Create a cron schedule for a recurring administrative task, save it in crontab and verify the entry.
15. Configure SSH key-based authentication between two systems and verify passwordless login.
16. Change the SSH server listening port, update the firewall accordingly, restart SSH and verify remote access.
17. Install and configure Apache HTTPD, start and enable the service, create a web page and test it locally.
18. Configure Apache access through firewalld and verify the web page from another client system.
19. Troubleshoot an Apache service that is installed but not accessible by checking service status, listening ports and firewall rules.
20. Perform a basic Linux system/network verification using uptime, free, df, ip addr show, ping and relevant process/service commands.

PART D – 6 MARK PRACTICAL QUESTIONS

1. Create a complete user/group setup for a project: create users and a group, assign primary/supplementary groups, configure home directories and passwords, and verify account details.
2. Implement a secure shared directory for a team using ownership, group permissions, setgid/sticky-bit behavior, and ACL. Demonstrate access for different users.
3. Perform a complete file-management task: create directories and files, copy/move/remove files, create hard and symbolic links, archive the data with TAR and compress it.
4. Manage a package lifecycle using DNF: search for a package, inspect its information, install it, verify the installation, review transaction history and remove it.
5. Configure a network connection with nmcli, assign IPv4 addressing information, activate the connection, verify the configuration and test connectivity.
6. Create and manage processes: run a command in the background, inspect it using ps/top, change its priority and terminate it. Demonstrate jobs, bg and fg.
7. Perform complete systemd service administration for a selected service, including start/stop, status checking, enable/disable at boot and verification.
8. Configure a firewall policy using firewalld: allow a required service and port permanently, reload the firewall, list active rules and verify connectivity.
9. Create and verify a scheduled administrative task using cron. Explain the six cron fields and demonstrate the configured schedule.
10. Configure SSH key-based authentication between client and server. Generate keys, transfer the public key, test login and verify the default SSH port.

- 11.** Secure SSH by changing the server port in `sshd_config`, updating `firewalld`, restarting the service and testing the new connection.
- 12.** Deploy an Apache HTTPD web server from start to finish: install the package, start and enable the service, create a test web page, configure the firewall and verify access.
- 13.** Troubleshoot a web server that cannot be reached remotely. Check Apache status, listening ports, firewall configuration, service configuration and client connectivity.
- 14.** Build a small Linux administration scenario combining user creation, group permissions, ACL, package installation, service management and firewall configuration.
- 15.** Perform a complete system administration verification: inspect disk/memory/CPU, network configuration, running processes, enabled services and firewall rules, then document the commands used.
- 16.** Create a practical task that uses DNF, `systemd`, `firewalld` and Apache together to deploy a basic web server and make it accessible to a client system.